



Cite this: *J. Mater. Chem. B*, 2023,
11, 2904

Dual functionalized brush copolymers as versatile antifouling coatings†

Jihyo Kil,^{‡a} Rafia Tasnim Rahman,^{‡b} Wenxuan Wang,^a Saehan Choi,^b
Yoon Sung Nam^{✉b} and Sheng Li^{✉*a}

Polymer coatings containing both fouling-resistant and fouling-release components have been reported to show synergistic antifouling properties. However, it remains unclear how the polymer composition influences the antifouling performance, particularly regarding foulants of different sizes and biological natures. Herein, we prepare dual functionalized brush copolymers containing fouling-resistant poly(ethylene glycol) (PEG) and fouling-release polydimethylsiloxane (PDMS) and examine their antifouling performances against different biofoulants. We utilize poly(pentafluorophenyl acrylate) (PPFPA) as a reactive precursor polymer and graft amine-functionalized PEG and PDMS side chains to create PPFPA-*g*-PEG-*g*-PDMS brush copolymers of systematically varying compositions. The copolymer films spin-coated on silicon wafers exhibit surface heterogeneity that can be correlated well with the bulk composition of the copolymer. When the copolymer-coated surfaces are examined against protein (human serum albumin and bovine serum albumin) adsorption and cell (lung cancer cells and microalgae) adhesion, they are found to perform better than the homopolymers. The enhanced antifouling properties are attributed to the copolymers having a PEG-rich top layer and a PEG/PDMS mixed bottom layer that work synergistically to resist biofoulant attachment. Furthermore, the composition of the best-performing copolymer is different for different foulants, with PPFPA-*g*-PEG₃₉-*g*-PDMS₄₆ exhibiting the best antifouling properties against proteins and PPFPA-*g*-PEG₅₄-*g*-PDMS₃₀ exhibiting the best antifouling properties against cells. We explain this difference by considering the changes in the length scale of the surface heterogeneity in relation to the foulant sizes.

Received 18th November 2022,
Accepted 16th February 2023

DOI: 10.1039/d2tb02522a

rsc.li/materials-b

Introduction

Antifouling surfaces have important implications for many industrial applications. The initial demand comes from the treatment of submerged structures and vessels.^{1–3} For the marine industry, the build-up of biofoulants on vessel surfaces can reduce their life span and fuel efficiency;^{4–6} thus, appropriate antifouling treatment is essential to vessel maintenance and operation. Another area showing increasing antifouling demand is coatings for biomedical implants and biosensors.^{7,8} These devices often require surface treatment to counter contamination by proteins and cells, and the build-up of foulants not only decreases the device efficiency but also poses serious health

risks.⁹ In efforts to prepare surfaces that can counter the settlement and adhesion of biofoulants, various approaches such as using biocide-containing paints^{10,11} and fabricating surfaces with micro- and nano-sized features^{12,13} have been explored. Among them, the use of polymer-based antifouling coatings stands out as an attractive option due to their low toxicity, large-scale applicability, and high efficiency.^{14,15} The polymer coatings can be generally categorized into two classes. One class is hydrophilic polymers with fouling-resistant properties, with representative examples including poly(ethylene glycol) (PEG), polysaccharides, and zwitterionic polymers.^{16–18} These polymers prefer interaction with water over biofoulants, thus effectively reducing the water-surface interfacial energy and decreasing the driving force for nonspecific adsorption.¹⁹ The second class is hydrophobic polymers with fouling-release properties, with representative examples including polydimethylsiloxane (PDMS) and fluoropolymers.^{20,21} These polymers have low surface free energy and high elasticity, thus attached biofoulants can be easily removed by modest shear force.^{22,23}

While both types of polymer coatings show antifouling effects, their performance in countering biofoulant build-up is limited. Studies have indicated that amphiphilic copolymers

^a Department of Chemical and Biomolecular Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon 34141, Republic of Korea.

E-mail: shengli@kaist.ac.kr

^b Department of Materials Science and Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon 34141, Republic of Korea.

E-mail: yoonsung@kaist.ac.kr

† Electronic supplementary information (ESI) available. See DOI: <https://doi.org/10.1039/d2tb02522a>

‡ These authors contributed equally to this work.

containing both fouling-resistant and fouling-release segments exhibit synergistic effects leading to enhanced performances.^{24–29} Baxamusa and Gleason reported random copolymers prepared from hydroxyethylmethacrylate and perfluorodecylacrylate.²⁸ When copolymers of varying compositions were tested against protein adsorption, fewer proteins were found to adsorb on the copolymer surfaces than on the homopolymers, attributed to the compositional heterogeneities on the copolymer surfaces. In another example, Wenning *et al.* synthesized PDMS-based amphiphilic block copolymers consisting of oligo(ethylene glycol) and fluoroalkyl side groups of varying composition ratios.²⁹ While the amphiphilic block copolymers showed improved fouling-release properties against *Ulva linza*, a macroalga that favors polar surfaces, the same block copolymers were less effective in removing *Navicula incerta*, a unicellular diatom that favors nonpolar surfaces. This study suggests that the composition of amphiphilic copolymers for the best performance depends on the nature of the foulant.

In addition to the polymer composition, the chain architecture of the polymer also plays an important role. In particular, polymers with densely grafted brush-like chain structures have been reported to show enhanced antifouling properties.^{30–33} Ma *et al.* prepared brush polymer coatings by using surface-initiated atom transfer radical polymerization (ATRP) of oligo(ethylene glycol)methyl methacrylate.³² The polymer-coated surfaces with high chain density showed a positive correlation between the brush thickness and protein resistance. More recently, Xu *et al.* reported the synthesis of amphiphilic brush copolymers consisting of poly(2,2,2-trifluoroethylmethacrylate) and PEG side chains grafted to the same repeat unit along the polymer backbone.³³ The brush copolymer-coated surfaces exhibited both fouling-resistant and fouling-release properties, with the polymer sample of a high degree of backbone polymerization and long PEG side chains showing the best antifouling performance against protein and cells.

Despite the promising antifouling performances shown by a number of brush polymer systems, studies involving amphiphilic brush copolymers remain limited, and the influence of copolymer composition variations on their antifouling effects against different foulants remains unclear. Sundaram *et al.* attempted to investigate the antifouling properties of graft copolymers containing different ratios of PEG and PDMS side chains.³⁴ In this case, a modified block copolymer containing epoxide functional groups was utilized as a precursor polymer, and the overall grafting density remained low at about 40–50 mole percent. The resulting copolymers also showed no obvious correlation between the composition and antifouling effects. In this study, we prepare amphiphilic brush copolymers with densely grafted PEG and PDMS side chains by employing a reactive poly(pentafluorophenyl acrylate) (PPFPA) precursor polymer. The PPFPA polymer is synthesized *via* reversible addition-fragmentation chain transfer (RAFT) polymerization. By utilizing the reactive PFP ester groups, amine-functionalized PEG and PDMS chains are grafted onto the PPFPA backbone to yield brush copolymers of systematically varying compositions. The brush copolymers are spin-coated onto silicon wafers, and

the surfaces are tested against proteins, cancer cells, and marine cells. In all scenarios, the brush copolymers show better antifouling performances than the homopolymers. More importantly, the optimized brush composition for the best performance is found to be different for proteins and cells.

Experimental section

Materials

The pentafluorophenyl acrylate (PFPA) monomer was synthesized according to the procedures published before.³⁵ (3-Aminopropyl)triethoxysilane (APTES), 2-(dodecylthiocarbo-*no*thiylthio)-2-methylpropionic acid (DDMAT), anhydrous tetrahydrofuran (THF), anhydrous anisole, and anhydrous toluene solvents were purchased from Sigma Aldrich. 2,2'-Azobis(2-methylpropionitrile) (AIBN), purchased from DaeJung Chemicals, was recrystallized in methanol twice before use. Amino-terminated PEG (PEG-NH₂, $M_n = 1000 \text{ g mol}^{-1}$) was purchased from Biochempeg, and amino-terminated PDMS (PDMS-NH₂, $M_n = 1000 \text{ g mol}^{-1}$) was purchased from Polymer Source. Bovine serum albumin (BSA) and human serum albumin (HSA) were purchased from Sigma Aldrich, the green fluorescent protein-expressing human lung cancer cell line (A549-GFP) was purchased from the American Type Culture Collection (ATCC), and green microalga *Chlorella vulgaris* (*C. vulgaris*) was purchased from Freshwater Bioresources Culture Collection (FBCC). Phosphate buffered saline (PBS), RPMI 1640, fetal bovine serum (FBS), and penicillin-streptomycin (10 000 U mL⁻¹ penicillin and 10 µg mL⁻¹ streptomycin) were purchased from GIBCO/invitrogen, and the Bold's basal medium was purchased from PhytoTech Labs.

Synthesis of PPFPA

AIBN (1 mg, 0.0061 mmol), DDMAT (20.1 mg, 0.0551 mmol), and PFPA monomer (1.74 g, 7.29 mmol) were combined with 2.4 mL of anhydrous anisole in a Schlenk flask. The mixture was degassed using three freeze-pump-thaw cycles. The Schlenk flask was immersed in a preheated oil bath at 70 °C for 6 h. The reaction was terminated by placing the flask in cold water and exposing the reaction content to air. The reaction mixture was diluted with THF, and then precipitated in cold methanol twice. The precipitated polymer was collected and then dried in a vacuum oven at 50 °C overnight. ¹H NMR (400 MHz; CDCl₃; TMS) δ: 3.09 (1H, s), 2.59–1.85 (2H, m). SEC (PS standard) $M_n = 21 \text{ kg mol}^{-1}$; PDI = 1.1.

Synthesis of brush copolymers (PPFPA-g-PEG-g-PDMS)

To prepare PEG and PDMS dual substituted brush copolymers, PEG-NH₂ and PDMS-NH₂ of desired molar concentrations were mixed with PPFPA. In a typical experiment, PPFPA (100 mg) and PEG-NH₂ of the desired amount were separately dissolved in 5 mL of anhydrous THF. The two solutions were combined, purged with nitrogen gas, and then reacted at 50 °C for 4 h. The degree of PEG substitution was determined by ¹⁹F NMR spectra (376 MHz; CDCl₃). PDMS-NH₂ of the desired

amount was subsequently added to the reaction mixture and then reacted at 50 °C for 4 h under nitrogen. The degree of PDMS substitution was again determined by ^{19}F NMR. The copolymer sample was recovered by the removal of the solvent using a rotary evaporator, and then further dried in a vacuum oven at 50 °C overnight.

Preparation of copolymer coatings on Si wafers

APTES linkers were deposited on Si wafer surfaces by following the previously reported method.³⁶ The synthesized PPFPA-*g*-PEG-*g*-PDMS copolymers were dissolved in anhydrous toluene at a concentration of 1 wt%. The copolymer solutions were spin-coated onto APTES-treated Si wafer surfaces at 5000 rpm for 60 s (spin coater Model ACE-200, Dong Ah Tech). The copolymer-coated wafers were then annealed in a vacuum oven at 130 °C for 1 h. They were finally rinsed with toluene to remove any loosely bound polymers and then dried under nitrogen.

Characterization

^1H NMR and ^{19}F NMR spectra were analyzed in CDCl_3 using a Bruker Avance III HD spectrometer (9.4T, ^1H = 400 MHz, ^{19}F = 376 MHz) at room temperature. The tetramethylsilane (TMS) signal was used as a reference peak. Fourier-transform infrared (FTIR) spectra were acquired in the transmittance mode using a Nicolet iS50 FTIR spectrometer (Thermo Fisher Scientific Instrument). The precursor PPFPA molecular weight was determined by THF size exclusion chromatography (SEC, Agilent 1260 Infinity II). The instrument was equipped with two PLgel 5 μm Mixed-C columns connected in series and a refractive index (RI) detector (G7800A 1260 GPC/SEC MDS). THF was used as the mobile phase at 35 °C with 1 mL min^{-1} flow rate. Calibration was performed using EasiVial PS-M standards (370–364 000 g mol^{-1}). The PPFPA-*g*-PEG copolymer molecular weights were determined by DMF SEC (YL9100 Plus HPLC system). The instrument was equipped with a YL9170 differential RI detector and a column-heating module with an ASAHIPAK GF-1G 7B guard column and two ASAHIPAK GF-7M HQ columns connected in series. The columns were eluted with DMF containing 0.05 M LiBr at 50 °C and a flow rate of 0.6 mL min^{-1} . Calibration was performed using Shodex PS standards (580–1 740 000 g mol^{-1}). The thickness of the copolymer coating on Si wafer was measured by ellipsometry (Alpha SE, J. A. Woollam Co., Inc.) with a wavelength range of 380–900 nm (1.38–3.25 eV) at three incident angles (65°, 70°, and 75°) to the silicon wafer using a transparent film mode. For each sample, the thickness measurements were obtained at five different spots and then analyzed using the Complete EASE software. The static water contact angle measurements were obtained using a drop shape analyzer (DSA10-Mk2, Kruss) at room temperature. After placing the water droplet, each copolymer-coated surface was given 180 s to reach its equilibrium wetting condition, and then the contact angle measurement was recorded. The surface composition of the copolymer coating was analyzed by X-ray photoelectron spectroscopy (XPS, Sigma Probe, Thermo VG Scientific). The XPS spectra were

acquired using an Al K α source accelerated at 15 kV and a beam current of 6.70 mA. Each spectrum was corrected for sample charging, and the positions of the centroids were adjusted by setting the hydrocarbon signal to 284.8 eV. The depth profile of a selected coating sample was obtained in the positive ion mode by time-of-flight secondary ion mass spectrometry (TOF-SIMS, TOF-SIMS5, ION-TOF GmbH). The spectra were acquired using a Bi_3^{++} primary ion beam accelerated at 60 keV over an analysis area of 100 $\mu\text{m} \times 100 \mu\text{m}$. The polymer layer was sputtered by using an Ar gas cluster ion beam (cluster size 1250 atoms) accelerated at 2.5 keV. The surface morphologies of selective copolymer coatings were analyzed by atomic force microscopy (AFM, XE-100, Park systems). The height images were acquired in the non-contact mode with a scan size of 500 nm $\times$ 500 nm. The mean domain areas were calculated using Park systems XEI software and setting 0.5 nm as a threshold height.

Protein adsorption

BSA or HSA (400 μg) was dissolved in 0.01 M PBS (20 mL, pH 7.4). The PPFPA-*g*-PEG-*g*-PDMS-coated Si wafers were then immersed in the BSA or HSA solution (20 $\mu\text{g mL}^{-1}$). After incubation at 37 °C with shaking at 50 rpm for 8 h, the amount of BSA or HSA adsorbed on the surfaces was determined by the micro-BCA assay and quantified by comparing the measured optical density of protein against standard solutions of known protein concentrations.

Mammalian cell adhesion

A549-GFP cells were cultured at 37 °C under a humidified 5% CO_2 atmosphere in RPMI 1640 supplemented with 10 vol% FBS and 1 vol% penicillin-streptomycin. The culture medium was replaced once every 2 days. After confluence, the cells were subcultured following trypsinization. Each PPFPA-*g*-PEG-*g*-PDMS-coated surface was placed in a six-well culture plate. The cells were seeded on the plate at a density of 3×10^5 cells per well and then incubated for 24 h at 37 °C in RPMI 1640. After washing with PBS twice, the cells were fixed with 4 wt% paraformaldehyde at 4 °C for 15 min. The fluorescence images were obtained using a fluorescence microscope (Eclipse 80i, Nikon) with the following settings: an excitation filter at 450–490 nm, a dichroic mirror at 505 nm, and an emission filter at 520 nm. The number of cells was quantified using Image J software. For scanning electron microscopy (SEM, SU5000, Hitachi) observation, the fixed A549 cells were gradually dehydrated in a series of ethanol solutions (30, 50, 60, 70, 80, 90, and three times 100 vol% ethanol) for 10 min each. The dehydrated cells were lyophilized and coated with platinum at ~ 3 nm using a sputter coater (SCD005, BAL-TEC). To assess the cytotoxicity of the brush copolymers, A549 cells (1×10^4 cells per well) were transferred to 96 well plates in 100 μL of RPMI 1640 and attached overnight. The cells were treated with 50 $\mu\text{g mL}^{-1}$ of the brush copolymers, followed by incubation at 37 °C for 24 h. The same number of cells seeded without any addition of the polymer was used as a control, and a culture medium without cells was used as a blank. After 24 h

of incubation, 10 μL of CCK solutions (10 vol% in RPMI 1640) (Cell Counting Kit-8, DOJINDO Laboratories) were added to each well. The plate was further incubated at 37 $^{\circ}\text{C}$ for 1 h. The absorbance at 450 nm was recorded using a microplate reader.

Microalga settlement

C. vulgaris was inoculated in a 15 mL tube containing 10 mL Bold's basal media and then incubated at 20 $^{\circ}\text{C}$ with 3000 Lux continuous cool-white fluorescent light illumination. The lamp was switched on and off for 16 h and 8 h alternatively to simulate sunlight irradiation. The culture tube was hand-shaken three times daily during the incubation period to avoid algae sticking. After a 7 day growth period, the culture medium was combined with a fresh culture medium to obtain a final *C. vulgaris* concentration of 3×10^5 per well to be used as the test medium for the settle experiments. Each PPFPA-*g*-PEG-*g*-PDMS-coated surface was placed in a six-well culture plate. *C. vulgaris* medium (2 mL) was added to each well and then incubated for 7 days. The fluorescence images were obtained after 3 days and 7 days with a fluorescence microscope with the following settings: an excitation filter at 620–660 nm, a dichroic mirror at 660 nm, and an emission filter at 700 nm. The settlement of *C. vulgaris* was determined by analyzing ten visual fields using Image J software. For SEM observation, the fixed cells were gradually dehydrated in a series of ethanol solutions (30, 50, 60, 70, 80, 90, and three times 100 vol% ethanol) for 10 min each. The dehydrated cells were lyophilized and coated with platinum at ~ 3 nm using a sputter coater.

Results and discussion

Synthesis of PPFPA-*g*-PEG-*g*-PDMS from a reactive precursor

The reactive precursor polymer PPFPA was synthesized by RAFT polymerization using DDMAT as the chain transfer agent,

following previously reported chemistry.^{35,36} The SEC elution profile of PPFPA (Fig. S1, ESI[†]) showed a narrow molecular weight distribution with a polystyrene (PS) equivalent M_n of 21 kg mol⁻¹. The PFP ester groups are known to have high reactivity toward amine groups;^{37,38} thus, by conducting post-polymerization substitution reactions, amino-terminated PEG and PDMS can be grafted onto PPFPA to create brush copolymers of the structure PPFPA-*g*-PEG-*g*-PDMS (Fig. 1a). In this study, we prepared brush copolymers with well-quantified side chain compositions by conducting the substitution reactions in sequence. The PPFPA precursor was first reacted with PEG-NH₂ ($M_n = 1$ kg mol⁻¹) in THF at 50 $^{\circ}\text{C}$ for 4 h to give PPFPA-*g*-PEG. The reaction mixture was then further reacted with PDMS-NH₂ ($M_n = 1$ kg mol⁻¹) under the same reaction conditions to yield the desired PPFPA-*g*-PEG-*g*-PDMS. By varying the feed concentrations of PEG-NH₂ and PDMS-NH₂, a total of seven brush copolymers with a similar total degree of side chain substitution but variable PEG and PDMS contents in the range of 0–80% were prepared (Table 1).

The formation of PPFPA-*g*-PEG by the reaction between PPFPA and PEG-NH₂ was first examined. The FTIR spectra of PPFPA-*g*-PEG prepared with different amounts of PEG-NH₂ are shown in Fig. 2a. As the feed concentration of PEG-NH₂ increased, the absorption band at 1780 cm⁻¹ corresponding to the ester group of PFP decreased in intensity, and a broad absorption band from 1690 to 1625 cm⁻¹ corresponding to the amide C=O stretch appeared and then increased in intensity. The amide band has been reported to be influenced by its hydrogen bonding with water and the neighboring functional groups.^{39–41} The peak at 1670 and 1650 cm⁻¹ may be attributed to free amide C=O and amide C=O hydrogen bonded to NH, respectively. The quantitative mole percent of PEG substitution was determined by ¹⁹F NMR (Fig. 2b). In the ¹⁹F NMR spectra, the three broad peaks in the range of –150 to –163 ppm are

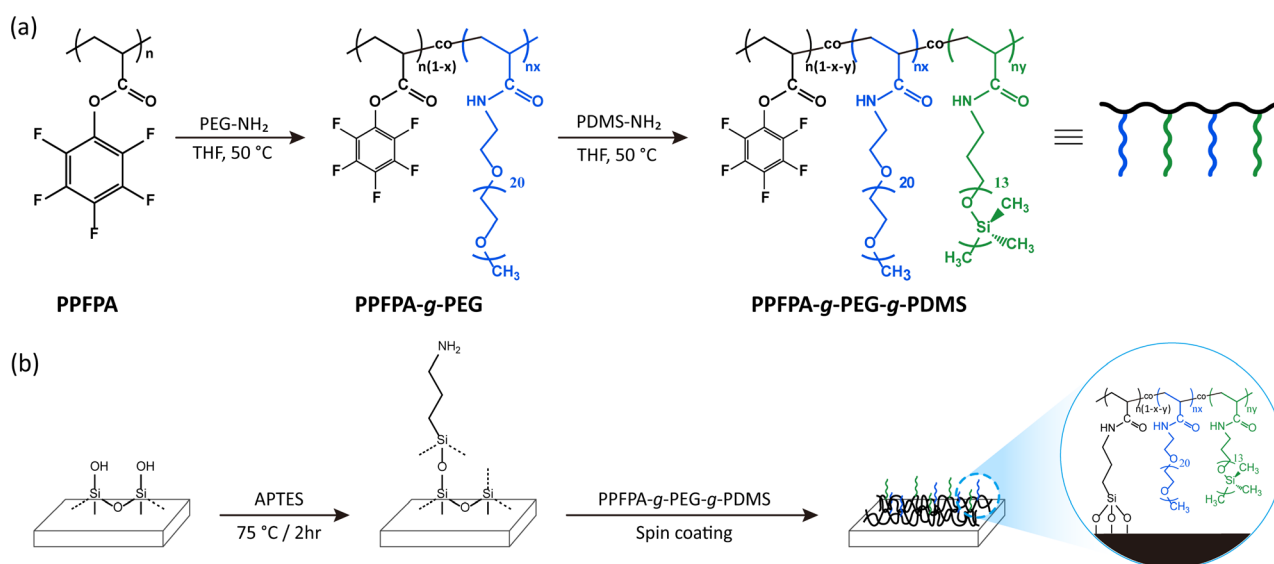


Fig. 1 (a) Synthesis of PPFPA-*g*-PEG-*g*-PDMS brush copolymers from a reactive precursor. (b) Schematics for the preparation of PPFPA-*g*-PEG-*g*-PDMS-coated Si wafer by using APTES as a linker.

Table 1 Characterization of PPFPA-*g*-PEG-*g*-PDMS brush copolymers

Sample		Feed concentration ^a		Brush copolymer		
Name	Code	PEG-NH ₂ (mol%)	PDMS-NH ₂ (mol%)	PEG ^b (mol%)	PDMS ^b (mol%)	M _n ^c (kg mol ⁻¹)
PPFPA- <i>g</i> -PEG ₀ - <i>g</i> -PDMS ₈₃	0-83	0	105	0	83	76
PPFPA- <i>g</i> -PEG ₇ - <i>g</i> -PDMS ₇₃	7-73	10	95	7	73	74
PPFPA- <i>g</i> -PEG ₂₇ - <i>g</i> -PDMS ₅₇	27-57	30	75	27	57	77
PPFPA- <i>g</i> -PEG ₃₉ - <i>g</i> -PDMS ₄₆	39-46	50	55	39	46	78
PPFPA- <i>g</i> -PEG ₅₄ - <i>g</i> -PDMS ₃₀	54-30	70	40	54	30	77
PPFPA- <i>g</i> -PEG ₇₁ - <i>g</i> -PDMS ₁₂	71-12	90	20	71	12	76
PPFPA- <i>g</i> -PEG ₇₈ - <i>g</i> -PDMS ₀	78-0	100	0	78	0	73

^a Calculated based on the number of PFFA repeat units. ^b Determined by ¹⁹F NMR in CDCl₃. ^c Calculated by SEC-measured PFFPA molecular weight and ¹⁹F NMR-measured mole percent of PEG and PDMS substitution.

associated with the PFP units remaining on the polymer, and the three sharp peaks in the range of −163 to −172 ppm are associated with the released PFP units due to new amide bond formation.⁴² By comparing the integrated peak ratios, the mole percent of PEG substitution for the copolymer samples was calculated to be in the range of 0 to 78% (Table 1). We designated the copolymers as PPFPA-*g*-PEG_{*x*}, where *x* denoted the mole percent of PEG substitution. The molecular weights of the copolymers were further examined by SEC using DMF as an

eluent. The SEC elution profiles (Fig. S2, ESI†) confirmed an increase in molecular weight as the degree of PEG substitution increased.

The PPFPA-*g*-PEG copolymers were further reacted with PDMS-NH₂ to yield the dual substituted brush copolymers PPFPA-*g*-PEG-*g*-PDMS. We aimed to prepare brush copolymers of the same size, and thus the feed concentration of PDMS-NH₂ was adjusted such that the overall degree of side chain substitution was similar for all the copolymers. The grafting of

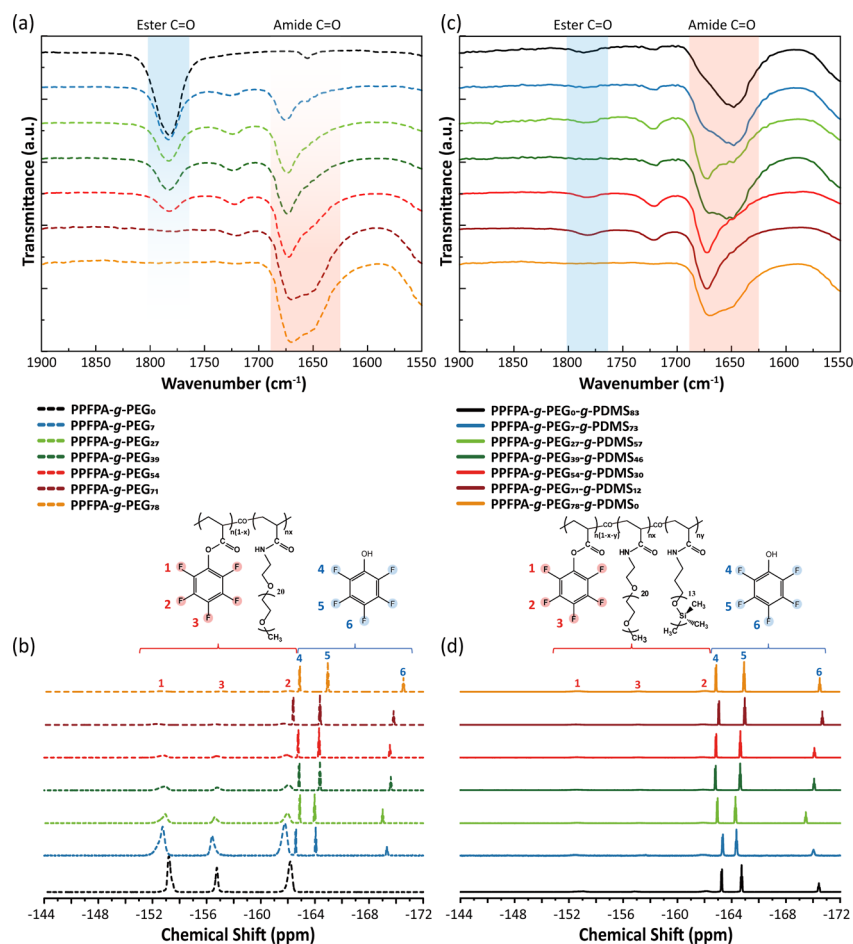


Fig. 2 FTIR (a) and ¹⁹F NMR (b) spectra of PPFPA-*g*-PEG copolymers. FTIR (c) and ¹⁹F NMR (d) spectra of PPFPA-*g*-PEG-*g*-PDMS copolymers.

PDMS to PPFPA was again confirmed by FTIR (Fig. 2c). All seven copolymer samples showed the amide functional group to be dominant, supporting the formation of densely grafted brush copolymers as a result of the successful substitution reaction. The quantitative total mole percent of side chain substitution was determined by ^{19}F NMR by following the same peak assignment as described before (Fig. 2d). All the copolymer samples were confirmed to have a similar total degree of side chain substitution of around 80%. The degree of PDMS substitution was also calculated by subtracting the known mole percent of PEG chains from the total mole percent of side chains. The calculated degree of PDMS substitution was in the range of 0 to 83% (Table 1). Note the 20% remaining PFP units were unreacted by design, as they would be later utilized for grafting the copolymers onto APTES-treated Si wafer surfaces. The final brush copolymers were designated as PPFPA-*g*-PEG_{*x*}-*g*-PDMS_{*y*}, where *y* denoted the mole percent of PDMS substitution. For ease of notation, the brush copolymer samples were also given a shorthand code of *x*-*y* (e.g., PPFPA-*g*-PEG₃₉-*g*-PDMS₄₆ = 39-46). The SEC elution profiles of PPFPA-*g*-PEG-*g*-PDMS were not obtained as the samples with high PDMS contents were not soluble in DMF. Instead, the final polymer *M_n* values were estimated based on the SEC-determined PPFPA molecular weight and the ^{19}F NMR-determined degree of side chain substitution. The detailed characterization data are summarized in Table 1.

Characterization of copolymer coatings

The synthesized PPFPA-*g*-PEG-*g*-PDMS brush copolymers with different PEG and PDMS side chain compositions were coated onto Si wafers by using APTES as a linker. The coating process is illustrated in Fig. 1b. The APTES linker molecules were first deposited onto Si wafer surfaces following the previously reported procedures.³⁶ The brush copolymers were dissolved in toluene at a concentration of 1 wt%, and then spin-coated onto APTES-treated surfaces. The polymer coatings were finally annealed at 130 °C for 1 h to promote the reaction between the terminal amine groups on APTES and the remaining PFP units (~20%) on the brush copolymers to give covalently grafted polymer coatings on the Si wafer surfaces. The method was followed to prepare polymer coatings from all seven brush copolymers. The resulting coating thickness was determined by ellipsometry. The total coating thickness was 11–26 nm, and the copolymer layer thickness was estimated to be 4–8 nm. A representative set of copolymer coatings with detailed thickness analysis is given in Table S1 (ESI†). The surface characteristics of the polymer coatings were examined by static water contact angles and XPS. Previous reports indicated that the surface properties of the polymer coatings could be different before and after water immersion, and the latter was more relevant to antifouling performance.^{43,44} Therefore, we treated the polymer-coated Si wafers by immersion in water for 12 h prior to conducting the surface characterization. The static water contact angles exhibited by the different polymer-coated surfaces are presented in Fig. 3a. For samples PPFPA-*g*-PEG₀-*g*-PDMS₈₃ and PPFPA-*g*-PEG₇₈-*g*-PDMS₀, the measured water contact angles were

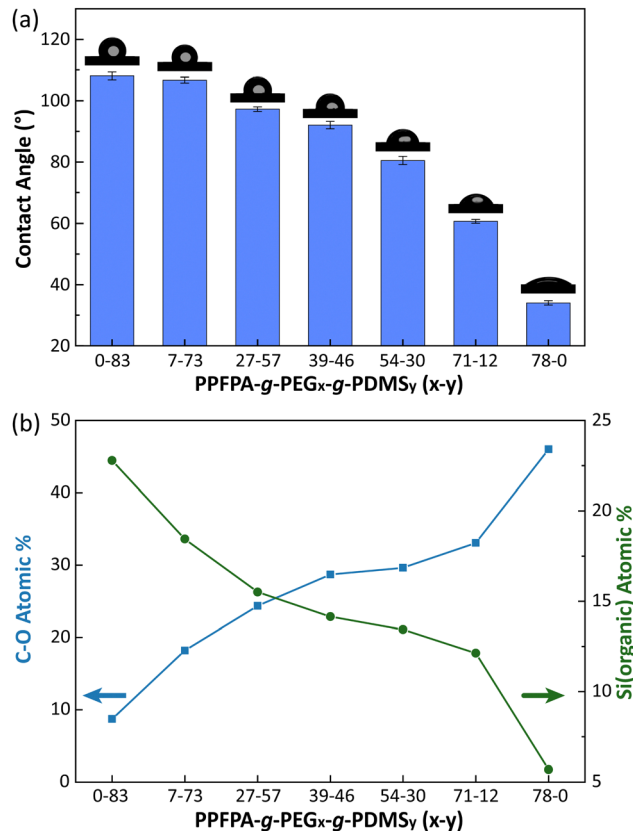


Fig. 3 (a) Static water contact angles and (b) XPS measured atomic percentages of C-O and Si(organic) for different PPFPA-*g*-PEG-*g*-PDMS coatings.

108° and 34°, respectively, in agreement with their hydrophobic and hydrophilic nature.^{45,46} For the copolymer coatings, as the PEG content increased, the water contact angle value decreased from 107° for PPFPA-*g*-PEG₇-*g*-PDMS₇₃ to 61° for PPFPA-*g*-PEG₇₁-*g*-PDMS₁₂. This result confirmed that the copolymers indeed exhibited different hydrophilicity in direct correlation with their compositions.

The surface compositions of the water pre-treated coating surfaces were further examined by XPS. The atomic percent of carbon, oxygen, and silicon were analyzed, and the detailed XPS spectra for the seven copolymer-coated surfaces are shown in Fig. S3–S9 (ESI†). The C1s signal could be resolved into two overlapping peaks from C-C at 284.6 eV and C-O at ~286 eV. The C-O peak was mainly associated with PEG side chains, and thus its intensity was used to estimate the presence of PEG on the coating surface. The Si2p peak mainly consisted of a Si(organic) peak at ~102 eV, attributed to the PDMS side chains; its intensity was used to estimate the presence of PDMS on the coating surface. Some samples also showed a contribution from Si(elemental) at 99.4 eV. This was due to the thin coating thickness and consequently the detection of some elemental Si from the underlying Si substrate. A comparison of the calculated atomic percentages of C-O and Si(organic) for the different polymer-coated surfaces is shown in Fig. 3b. The atomic percent of C-O increased from 8.7% to 46.0% as the

PEG side chain content of the copolymer coatings increased; the atomic percent of Si(organic) decreased from 22.8% to 5.7% as the PDMS side chain content decreased. The results indicated that the copolymer-coated surfaces were compositionally heterogeneous with both PEG and PDMS. The observed trend in the surface composition of the different coatings was also consistent with the bulk copolymer composition. We further compared the experimentally measured atomic percentages of C–O and Si(organic) to the theoretical values calculated based on the bulk composition (Table S2, ESI†). While the reported values for Si(organic) were found to be comparable, the experimentally determined C–O values were higher than the bulk theoretical values. These data further suggested that while the copolymer surfaces retained bulk composition heterogeneity, they exhibited a preference for the hydrophilic PEG chains.

Antifouling performance of copolymer coatings

We first examined the copolymer coatings against protein adsorption. The proteins BSA and HSA were chosen as their nonspecific adsorption on biomedical devices can cause serious health problems such as thrombus, fibrosis, and scar tissue formation.⁴⁷ Additionally, the adsorption of serum proteins is reported to facilitate cell adhesion, and thus serum albumin (SA) adsorption is often correlated with cell adhesion behavior.⁴⁸ Both BSA and HSA were dissolved in PBS at a concentration of $20 \mu\text{g mL}^{-1}$ and at pH 7.4. Note that the isoelectric point of SA is approximately 4.8, and it is negatively charged at near neutral pH.⁴⁹ The copolymer-coated Si wafers were immersed in the BSA or HSA solution, and then incubated at 37°C with shaking for 8 h. The total number of proteins adsorbed on each surface was quantified using a micro-BCA assay, and the results for HSA and BSA are shown in Fig. 4 and Fig. S10 (ESI†), respectively. For HSA adsorption, the PPFP-g-PEG₀-g-PDMS₈₃-coated surface showed a similar amount of adsorbed proteins as the bare Si surface. The PEG-substituted polymer (PPFP-g-PEG₇₈-g-PDMS₀) coating showed decreased protein adsorption, in agreement with previous reports;⁵⁰ however, the overall amount of HSA detected was still high at $58.7 \text{ pmol cm}^{-2}$. The copolymers containing both PEG and PDMS side chains showed much improved protein resistance in comparison to either PPFP-g-PEG₀-g-PDMS₈₃ or PPFP-g-PEG₇₈-g-PDMS₀, confirming the synergistic effect of having fouling-resistant and fouling-release segments on enhancing antifouling performance. More importantly, the degree of protein adsorption was found to be dependent on the composition of the copolymer coating, with the surface coated with PPFP-g-PEG₃₉-g-PEG₄₆ showing the least amount of adsorbed proteins at 8.8 pmol cm^{-2} , corresponding to a 90% decrease in protein adsorption in comparison to the bare Si surface. The adsorption behavior of BSA on the copolymer-coated surfaces followed a trend similar to that observed for HSA. The best-performing copolymer-coating was again confirmed to be PPFP-g-PEG₃₉-g-PEG₄₆, exhibiting an 89% decrease in BSA adsorption in comparison to the bare Si surface.

The antifouling performance of the copolymer coatings was further examined in cell adhesion experiments. Two types of

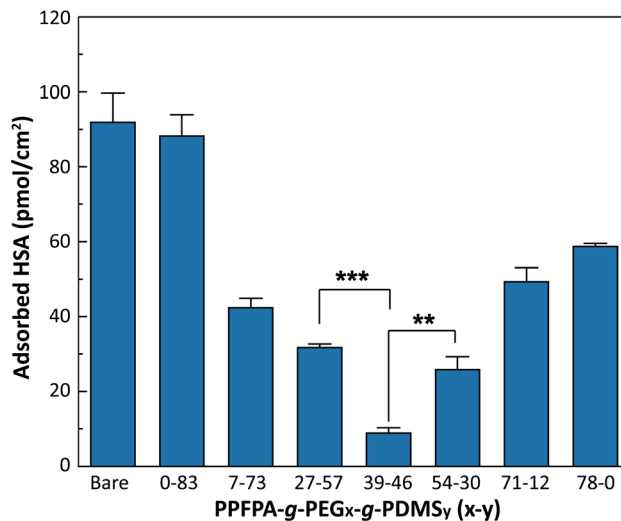


Fig. 4 Quantified average numbers of HSA adsorbed on the bare Si surface and PPFP-g-PEG-g-PDMS copolymer-coated Si surfaces ($n = 3$, and the error bars denote standard deviation). ** and *** represent p -values less than 0.01 and 0.001, respectively.

very different cells were selected, one representing mammalian cells and the other representing simple marine organisms. For the choice of mammalian cells, we used A549-GFP, an adenocarcinoma human lung epithelial cell genetically engineered to express GFP.⁵¹ The mechanism of A549 adhesion is well reported, involving a process in which the cells attach to each other and the surroundings through the cell-adhesion molecules (CAMs) on the cell surfaces.⁵² To examine the adhesion behavior, the A549-GFP cells were seeded at a density of 3×10^5 cells per well onto the various copolymer-coated Si wafer surfaces, followed by incubation at 37°C in RPMI 1640 for 24 h. The cells were then washed with PBS and fixed with 4% paraformaldehyde. The number of cells attached to the surfaces was visualized by green fluorescence microscopy. Representative images from the different copolymer-coated surfaces are shown in Fig. 5a, and the quantified number of attached cells is presented in Fig. 5b. Compared to the bare Si wafer surface, showing 630 cells per cm^2 , the PDMS-substituted (PPFP-g-PEG₀-g-PDMS₈₃) and PEG-substituted (PPFP-g-PEG₇₈-g-PDMS₀) polymer coatings showed a lower number of cells at 537 cells per cm^2 and 492 cells per cm^2 , respectively. This observation agreed with previous studies, which reported PEG or PDMS treated surfaces to have cell anti-adhesion abilities.^{53,54} The number of cells adhered to surfaces coated with PEG and PDMS dual-substituted copolymers was found to be further decreased. Among the different copolymer compositions examined, PPFP-g-PEG₅₄-g-PDMS₃₀ showed the least number of attached A549-GFP cells at 181 cells per cm^2 , representing an almost 4-fold drop in cell adhesion in comparison to the bare Si surface. It is interesting to note that in contrast to the copolymer composition for best protein resistance, cell adhesion is minimized at a different copolymer composition. We further examined the cytotoxicity of the brush copolymers by performing the CCK-8 assay (Fig. S11, ESI†). In

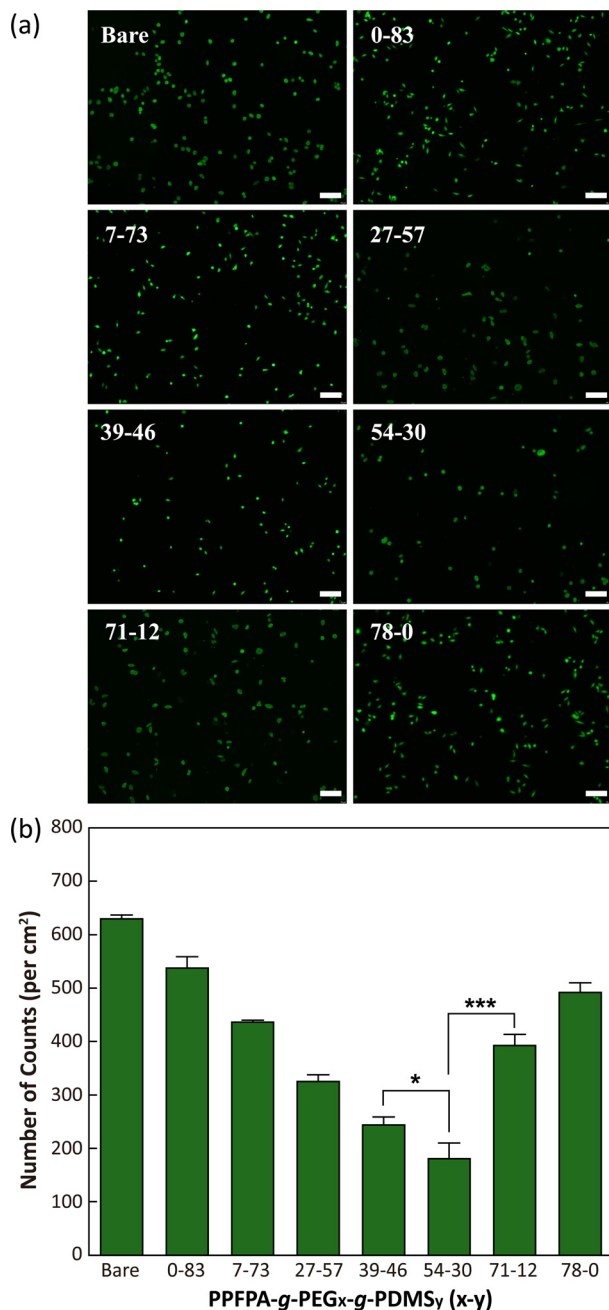


Fig. 5 (a) Fluorescence microscopy images of A549-GFP cells adhered to the bare Si surface and PPFPA-g-PEG-g-PDMS copolymer-coated Si surfaces after 24 h of incubation (scale bar: 200 μ m). (b) Quantified average numbers of adhered cells ($n = 3$, and the error bars denote standard deviation). * and *** represent p -values less than 0.05 and 0.001, respectively.

comparison to the control sample without any polymer addition, the cells incubated with the brush copolymers showed similar viability, confirming that the observed enhancement in antifouling behavior was not due to polymer toxicity induced cell death.

The second type of cells used to test the antifouling performance of the copolymer coatings was *C. vulgaris*, a species of green microalgae commonly reported as a representative marine foulant. The settlement of *C. vulgaris* follows a mechanism different from that of cancer cells. As *C. vulgaris* cannot

settle directly onto a surface, it adopts a process in which a biofilm composed of water and organic matters is first formed within 1–2 days, and then the larvae of the marine organism adhere, utilizing the biofilm as a medium for attachment as well as a nutrient source for growth.⁵⁵ To assess the settlement behavior, *C. vulgaris* (3×10^5 cells per well) were seeded onto the copolymer-coated Si wafers with 2 mL of culture medium. We tracked the settlement of *C. vulgaris* at two stages: first after 3 days, representing the initial phase of larvae deposition and attachment; second after 7 days, representing the cell's long-term growth and settlement behavior. The attachment of *C. vulgaris* to the different copolymer-coated surfaces was again visualized by fluorescence microscopy. Representative images along with analyzed cell counts after 3 days and 7 days are shown in Fig. S12 (ESI[†]) and Fig. 6, respectively. The 3 day and 7 day cell attachment behavior on the various surfaces followed a similar trend, with the 7 day cell counts showing an overall decrease in number compared to the 3 day cell counts, attributed to the detachment of some cells during the transition period. Based on the 7 day cell attachment data shown in Fig. 6, the bare Si surface showed a high *C. vulgaris* count of 14 253 cells per cm^2 , indicating a strong preference for *C. vulgaris* to settle on bare Si surfaces due to the fast formation of biofilms. The copolymer-coated surfaces showed fewer numbers of attached *C. vulgaris*, again with the PEG and PDMS dual-substituted copolymer coatings showing better antifouling performances than the PEG or PDMS single-substituted polymer coatings. The copolymer surface exhibiting the best antifouling performance against *C. vulgaris* was PPFPA-g-PEG₅₄-g-PDMS₃₀, with 480 cells per cm^2 after 7 days of incubation, corresponding to an impressive 30-fold reduction in cell counts in comparison to the bare Si wafer.

To better understand the attachment of the cells on the different polymer-coated surfaces, we further examined the cell morphology on three representative copolymer-coated surfaces by SEM. As shown in Fig. 7a, A549 cells on the PPFPA-g-PEG₇-g-PDMS₇₃-coated surface formed an extended morphology, indicating a relatively extensive spreading of the cell due to favorable surface interactions. We also quantified the relative degree of cellular membrane adhesion by calculating the ratio of the peripheral area to the cell body area. For A549 on PPFPA-g-PEG₇-g-PDMS₇₃, the ratio was determined to be 3.4. As the composition of the copolymer coating changed to PPFPA-g-PEG₃₉-g-PDMS₄₆ and PPFPA-g-PEG₅₄-g-PDMS₃₀, the calculated ratio decreased to 1.9 and 1.4, respectively. The same observation was also made for *C. vulgaris* (Fig. 7b), where the ratio of the area decreased in the order of 3.3 (PPFPA-g-PEG₇-g-PDMS₇₃), 2.1 (PPFPA-g-PEG₃₉-g-PDMS₄₆), and 1.3 (PPFPA-g-PEG₅₄-g-PDMS₃₀). This analysis showed that despite the very different attachment mechanisms, both A549 and *C. vulgaris* showed the least preference for PPFPA-g-PEG₅₄-g-PDMS₃₀-coated surfaces, which then explained the observed least number of attached cells on the same surfaces.

Comparing the antifouling data across proteins and cells, we found that the copolymer coatings containing both PEG and PDMS side chains performed better in all cases. We further

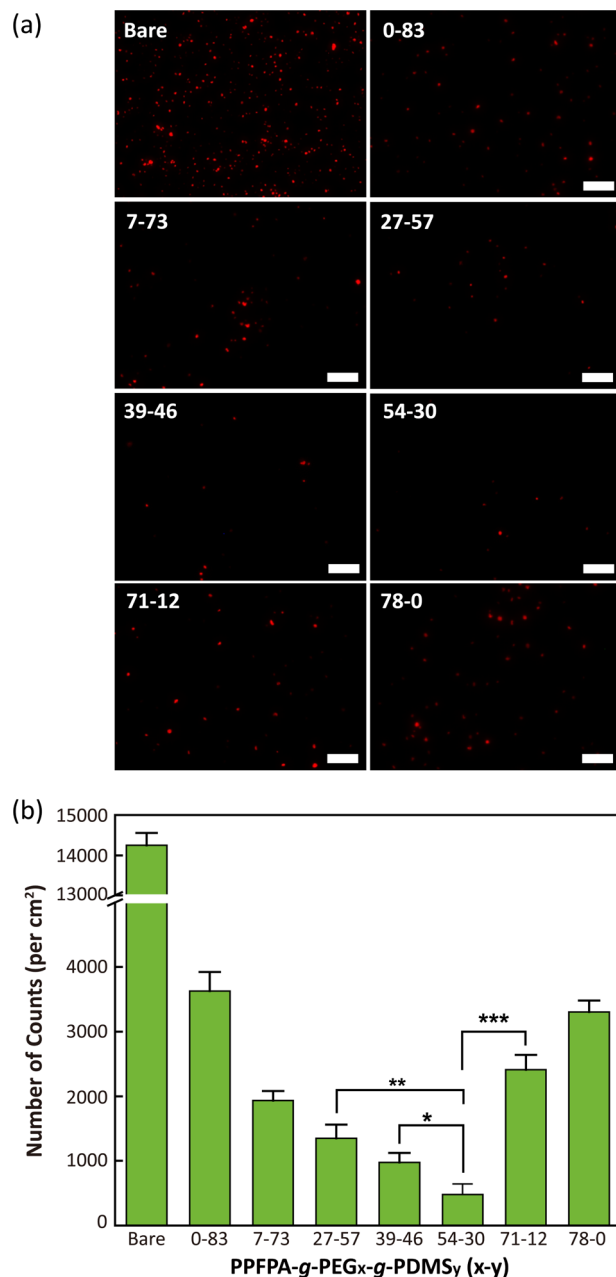


Fig. 6 (a) Fluorescence microscopy images of *C. vulgaris* settled on the bare Si surface and PPFP-g-PEG-g-PDMS copolymer-coated Si surfaces after 7 days (scale bar: 20 μm). (b) Quantified average numbers of attached *C. vulgaris* ($n = 3$, and the error bars denote standard deviation). * indicates a p -value of 0.05. ** indicates a p -value of 0.01. *** indicates a p -value of 0.001.

conducted depth profiling for one of the best-performing coating samples, PPFP-g-PEG₅₄-g-PDMS₃₀, by TOF-SIMS (Fig. 8a). In the positive ion mode, the measured secondary ion intensities of m/z 147 ($\text{C}_7\text{H}_{15}\text{O}_3^+$), m/z 221 ($\text{C}_7\text{H}_{21}\text{O}_2\text{Si}_3^+$), and m/z 56 (Si_2^+) were assigned to PEG, PDMS, and the Si substrate, respectively. At low sputter times labeled as region I, the top layer of the coating was probed, showing PEG chains to be dominant in composition. This observation was in agreement with XPS results, and both indicated that the top surface of the

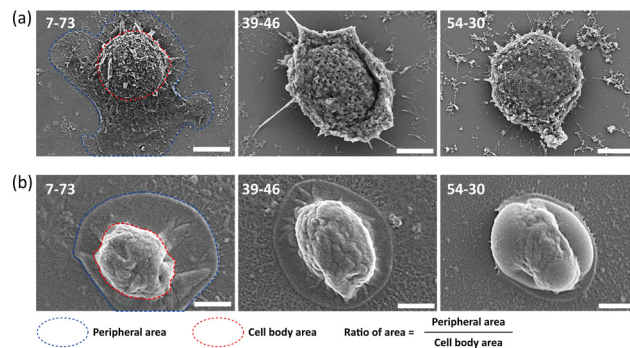


Fig. 7 SEM images of (a) A549 cells (scale bar: 10 μm) and (b) *C. vulgaris* (scale bar: 2 μm) on three different PPFP-g-PEG_x-g-PDMS_y (x - y) copolymer-coated surfaces. The ratio of the area is calculated by dividing the peripheral area (indicated in blue) by the cell body area (indicated in red).

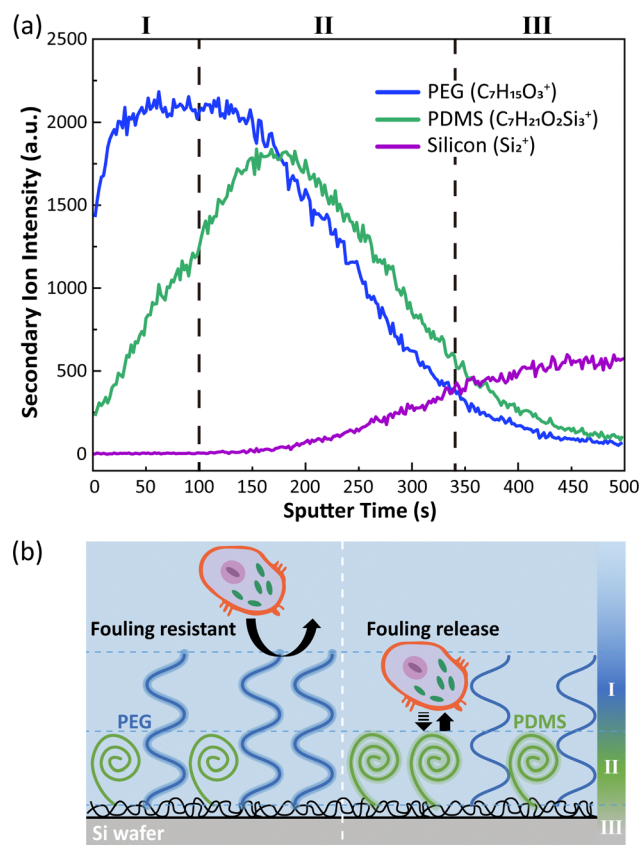


Fig. 8 (a) Depth profile of the PPFP-g-PEG₅₄-g-PDMS₃₀ copolymer coating on the Si substrate. (b) Schematic illustration of the heterogeneous surface morphology of the PPFP-g-PEG-g-PDMS copolymer coating and its antifouling properties.

polymer coating was rich in PEG. At longer sputter times labeled as region II, the characteristic secondary ion intensity for PDMS increased until its magnitude became comparable to that of the PEG intensity, suggesting that region II contained PEG and PDMS in coexistence. As the sputter time continuously

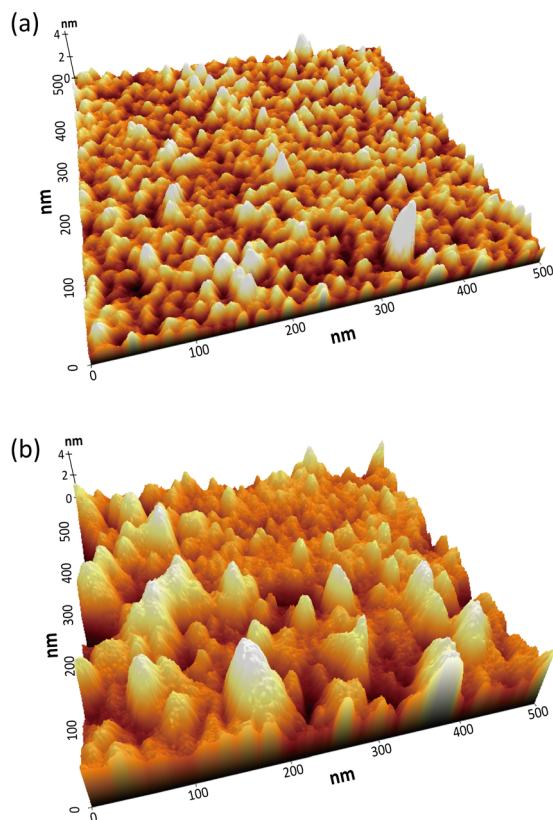


Fig. 9 AFM height images of (a) PPFPA-*g*-PEG₃₉-*g*-PDMS₄₆ and (b) PPFPA-*g*-PEG₅₄-*g*-PDMS₃₀ coated surfaces after incubation in water for 12 h and drying.

increased into region III, both the PEG and PDMS intensities diminished while the intensity of Si₂⁺ increased and became dominant. This transition signified that the polymer coating was sputtered through to reach the underlying Si substrate. Note that the depth profile showed broad transitions between regions, indicating that the layers had significant phase mixing. This observation was consistent with expectations as the PEG and PDMS side chains were randomly grafted onto the precursor polymer.

When the copolymer-coated substrates are exposed to an aqueous environment, the PEG side chains are stretched while the PDMS side chains are collapsed. This difference in chain conformation then creates a laterally heterogeneous surface morphology with a PEG-rich top layer and a mixed PEG/PDMS bottom layer, as illustrated in Fig. 8b. When foulants encounter the copolymer coatings, the PEG-rich top layer serves as the first line of defense, and the hydrophilicity of the polymer effectively repels the foulants and minimizes their adhesion to the coated surface. As some foulants inevitably penetrate through the PEG-rich top layer, the bottom mixed layer serves as the second line of defense. Here, the PEG and PDMS chains work in conjunction to facilitate the easy detachment of the foulants by utilizing the fouling-release property of PDMS. The surface heterogeneity enhanced antifouling behavior explains the superior performance of the copolymer coatings in comparison to the homopolymers.

Furthermore, we observed that the copolymer composition with the best antifouling performance was different depending on the foulants: PPFPA-*g*-PEG₃₉-*g*-PDMS₄₆ for small proteins vs. PPFPA-*g*-PEG₅₄-*g*-PDMS₃₀ for larger cells. The AFM height images of the PPFPA-*g*-PEG₃₉-*g*-PDMS₄₆-coated and PPFPA-*g*-PEG₅₄-*g*-PDMS₃₀-coated surfaces, following water pre-treatment, were obtained to evaluate the differences in their surface morphologies. As illustrated in Fig. 9, both copolymer-coated surfaces show complex domain morphologies but differ in their feature sizes. The peaks seen in the AFM images increased in size as the PEG content of the copolymer increased, indicating that the peaks could be associated with PEG-rich domains. We further estimated the sizes of the PEG-rich domains by setting a height threshold value of 0.5 nm and then calculating all the domain areas with heights above the threshold using imaging software (Fig. S13, ESI†). For the PPFPA-*g*-PEG₃₉-*g*-PDMS₄₆-coated surface, the PEG-rich domains were fairly regular in size distribution with a mean area of 147 nm². For the PPFPA-*g*-PEG₅₄-*g*-PDMS₃₀-coated surface, the PEG-rich domains showed a broad size distribution with a mean area of 738 nm². The results suggest that the differences in the copolymer composition lead to the formation of surface domain morphologies with different length scales, which in turn show disfavorable interactions with foulants of different sizes. This idea that the length scale of the surface heterogeneity should be correlated with the foulant size has been reported before.^{28,33,45,56} Most notably, Wooley and coworkers examined the antifouling properties of crosslinked polymer coatings prepared from hyperbranched fluoropolymers and linear PEG.^{45,57} They reported the copolymer network with 45% PEG to have the best antifouling properties against BSA, attributed to the small length scale of the surface domain morphology comparable to that of the protein. When larger marine foulants were tested, the settlement of the spores was found to be the lowest for the copolymer network with a higher 55% PEG content, although further experiments on spore removal under shear stress showed the copolymer with 45% PEG to have better spore removal performance. Overall, our results are in agreement with the earlier reports, and they support the argument that to effectively discourage foulant attachment to a surface, the polymer coating should have an optimized composition such that the resulting surface heterogeneity matches with the foulant size on the length scale.

Conclusions

We prepared brush copolymers containing fouling-resistant PEG and fouling-release PDMS side chains by utilizing a reactive PPFPA homopolymer precursor. By controlling the PEG and PDMS feed concentrations, brush copolymers with systematically variable PEG and PDMS contents were obtained. The resulting brush copolymers were then coated onto Si substrates and examined for their surface properties. The heterogeneous composition of the polymer coating was confirmed, where surface PEG and PDMS contents were well

correlated with bulk copolymer composition. We further examined the antifouling performance of the copolymer coating against proteins, cancer cells, and microalgae. In all cases, the dual-substituted copolymers were determined to show enhanced performances compared to the PEG or PDMS single-substituted polymers. Moreover, the copolymer composition for the best antifouling performance could be correlated with the foulant size, with the smaller proteins showing the least attachment to the PPFPA-*g*-PEG₃₉-*g*-PDMS₄₆-coated surface and the larger cells showing the least attachment to the PPFPA-*g*-PEG₅₄-*g*-PDMS₃₀-coated surface. We attributed the enhanced antifouling properties of the copolymers to two factors. First, due to the different extents of chain stretching in an aqueous environment, the copolymer coatings show a PEG-rich top layer and a mixed PEG/PDMS bottom layer. This chain arrangement creates two lines of defense that work effectively against foulant attachment. Second, the copolymer composition directly influences the length scale of the resulting surface domain morphology. The best antifouling performance is observed when the domain size is comparable to the foulant size. Overall, we present in this study a versatile reactive polymer platform to access brush copolymers of tunable side chain content and illustrate the importance of designing copolymer coatings with optimized composition and surface heterogeneity for antifouling against different biofoulants.

Author contributions

J. Kil and R. T. Rahman contributed to data curation, formal analysis, investigation, methodology, visualization, and writing of the original draft, review, and editing. W. Wang contributed to formal analysis, investigation and validation. S. Choi contributed to investigation and resources. Y. S. Nam contributed to supervision, project administration, funding acquisition, and review and editing of the manuscript. S. Li contributed to conceptualization, supervision, project administration, funding acquisition, and writing of the original draft, review, and editing. J. Kil and R. T. Rahman contributed equally to this work.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

This work was supported by the Korean Government Ministry of Science and ICT (NRF-2020R1C1C1007291).

References

- 1 X. Li, S. Li, X. Huang, Y. Chen, J. Cheng and A. Zhan, *Mar. Environ. Res.*, 2021, **170**, 105409.
- 2 H. Qiu, K. Feng, A. Gapeeva, K. Meurisch, S. Kaps, X. Li, L. Yu, Y. K. Mishra, R. Adelung and M. Baum, *Prog. Polym. Sci.*, 2022, 101516.
- 3 J. A. Callow and M. E. Callow, *Nat. Commun.*, 2011, **2**, 1–10.
- 4 M. E. Barry, E. C. Davidson, C. Zhang, A. L. Patterson, B. Yu, A. K. Leonardi, N. Duzen, K. Malaviya, J. L. Clarke and J. A. Finlay, *Macromolecules*, 2019, **52**, 1287–1295.
- 5 M. P. Schultz, *Biofouling*, 2007, **23**, 331–341.
- 6 M. Schultz, J. Walker, C. Steppe and K. Flack, *Biofouling*, 2015, **31**, 759–773.
- 7 M. Aghajani and F. Esmaeili, *J. Biomater. Sci., Polym. Ed.*, 2021, **32**, 1770–1789.
- 8 M. Alvarez-Paino, A. Munoz-Bonilla and M. Fernandez-Garcia, *Nanomaterials*, 2017, **7**, 48.
- 9 Z. He, X. Yang, N. Wang, L. Mu, J. Pan, X. Lan, H. Li and F. Deng, *Front. Bioeng. Biotechnol.*, 2021, **9**, 807357.
- 10 I. Omae, *Chem. Rev.*, 2003, **103**, 3431–3448.
- 11 N. Bellotti, C. Deya, B. del Amo and R. Romagnoli, *Ind. Eng. Chem. Res.*, 2010, **49**, 3386–3390.
- 12 J. Wang, Q. Du, J. Luan, X. Zhu and J. Pang, *Langmuir*, 2021, **37**, 7449–7456.
- 13 R. Mardosaite, A. Jurkeviciute and S. Rackauskas, *Cryst. Growth Des.*, 2021, **21**, 4765–4779.
- 14 M. S. Selim, M. Shenashen, S. A. El-Safty, S. Higazy, M. M. Selim, H. Isago and A. Elmarakbi, *Prog. Mater. Sci.*, 2017, **87**, 1–32.
- 15 S. Krishnan, C. J. Weinman and C. K. Ober, *J. Mater. Chem.*, 2008, **18**, 3405–3413.
- 16 S. Kim, T. Gim, Y. Jeong, J. H. Ryu and S. M. Kang, *ACS Appl. Mater. Interfaces*, 2017, **10**, 7626–7631.
- 17 J. Zhu and R. E. Marchant, *Biomacromolecules*, 2006, **7**, 1036–1041.
- 18 J. Yang, H. Chen, S. Xiao, M. Shen, F. Chen, P. Fan, M. Zhong and J. Zheng, *Langmuir*, 2015, **31**, 9125–9133.
- 19 R. Wang, H. Kreuzer and M. Grunze, *J. Phys. Chem. B*, 1997, **101**, 9767–9773.
- 20 E. Seo, J. W. Lee, D. Lee, M. R. Seong, G. H. Kim, D. S. Hwang and S. J. Lee, *Colloids Surf., B*, 2021, **207**, 112003.
- 21 J. C. Yarbrough, J. P. Rolland, J. M. DeSimone, M. E. Callow, J. A. Finlay and J. A. Callow, *Macromolecules*, 2006, **39**, 2521–2528.
- 22 A. Arrage, N. Vasishtha, D. Sundberg, G. Bausch, H. Vincent and D. White, *J. Ind. Microbiol.*, 1995, **15**, 277–282.
- 23 A. K. Leonardi and C. K. Ober, *Annu. Rev. Chem. Biomol. Eng.*, 2019, **10**, 241–264.
- 24 S. Krishnan, N. Wang, C. K. Ober, J. A. Finlay, M. E. Callow, J. A. Callow, A. Hexemer, K. E. Sohn, E. J. Kramer and D. A. Fischer, *Biomacromolecules*, 2006, **7**, 1449–1462.
- 25 X. Zhao, Y. Su, Y. Li, R. Zhang, J. Zhao and Z. Jiang, *J. Membr. Sci.*, 2014, **450**, 111–123.
- 26 P. Hu, Q. Xie, C. Ma and G. Zhang, *Langmuir*, 2020, **36**, 2170–2183.
- 27 H. Guo, C. Wen, S. Tian, X. Zhang, Y. Ma, X. Liu, J. Yang and L. Zhang, *ACS Appl. Mater. Interfaces*, 2021, **13**, 21051–21059.
- 28 S. H. Baxamusa and K. K. Gleason, *Adv. Funct. Mater.*, 2009, **19**, 3489–3496.
- 29 B. M. Wenning, E. Martinelli, S. Mieszkina, J. A. Finlay, D. Fischer, J. A. Callow, M. E. Callow, A. K. Leonardi, C. K. Ober and G. Galli, *ACS Appl. Mater. Interfaces*, 2017, **9**, 16505–16516.

- 30 W.-L. Chen, R. Cordero, H. Tran and C. K. Ober, *Macromolecules*, 2017, **50**, 4089–4113.
- 31 S. Gon and M. M. Santore, *Langmuir*, 2011, **27**, 15083–15091.
- 32 H. Ma, D. Li, X. Sheng, B. Zhao and A. Chilkoti, *Langmuir*, 2006, **22**, 3751–3756.
- 33 B. Xu, Y. Liu, X. Sun, J. Hu, P. Shi and X. Huang, *ACS Appl. Mater. Interfaces*, 2017, **9**, 16517–16523.
- 34 H. S. Sundaram, Y. Cho, M. D. Dimitriou, C. J. Weinman, J. A. Finlay, G. Cone, M. E. Callow, J. A. Callow, E. J. Kramer and C. K. Ober, *Biofouling*, 2011, **27**, 589–602.
- 35 J. Park, J. Nam, M. Seo and S. Li, *ACS Macro Lett.*, 2022, **11**, 468–474.
- 36 J. Ku, S. Kim, J. Park, T.-S. Kim, R. Kharbush, E.-C. Shin, K. Char, Y. Kim and S. Li, *Biomacromolecules*, 2020, **21**, 2440–2454.
- 37 C. Li, S. Feng, C. Li, Y. Sui, J. Shen, C. Huang, Y. Wu and W. Huang, *Adv. Funct. Mater.*, 2020, **30**, 2002163.
- 38 L. He, K. Szameit, H. Zhao, U. Hahn and P. Theato, *Biomacromolecules*, 2014, **15**, 3197–3205.
- 39 C.-W. Tu and S.-W. Kuo, *J. Polym. Res.*, 2014, **21**, 1–8.
- 40 Y. Maeda, T. Higuchi and I. Ikeda, *Langmuir*, 2000, **16**, 7503–7509.
- 41 B. Sun, Y. Lin, P. Wu and H. W. Siesler, *Macromolecules*, 2008, **41**, 1512–1520.
- 42 K. E. Doncom, C. F. Hansell, P. Theato and R. K. O'Reilly, *Polym. Chem.*, 2012, **3**, 3007–3015.
- 43 C. S. Gudipati, C. M. Greenlief, J. A. Johnson, P. Prayongpan and K. L. Wooley, *J. Polym. Sci., Part A: Polym. Chem.*, 2004, **42**, 6193–6208.
- 44 G. Galli and E. Martinelli, *Macromol. Rapid Commun.*, 2017, **38**, 1600704.
- 45 C. S. Gudipati, J. A. Finlay, J. A. Callow, M. E. Callow and K. L. Wooley, *Langmuir*, 2005, **21**, 3044–3053.
- 46 A. Tropmann, L. Tanguy, P. Koltay, R. Zengerle and L. Riegger, *Langmuir*, 2012, **28**, 8292–8295.
- 47 C. Zhao, L. Li, Q. Wang, Q. Yu and J. Zheng, *Langmuir*, 2011, **27**, 4906–4913.
- 48 D. T. H. Wassell and G. Embery, *Biomaterials*, 1996, **17**, 859–864.
- 49 J. Y. Park, J. Yeom, J. S. Kim, M. Lee, H. Lee and Y. S. Nam, *Macromol. Biosci.*, 2013, **13**, 1511–1519.
- 50 S. Lowe, N. M. O'Brien-Simpson and L. A. Connal, *Polym. Chem.*, 2015, **6**, 198–212.
- 51 C. Zhao, S. Setrerrahmane and H. Xu, *Oncol. Rep.*, 2015, **34**, 2126–2132.
- 52 B. M. Gumbiner, *Cell*, 1996, **84**, 345–357.
- 53 A. Zuchowska, P. Kwiatkowski, E. Jastrzebska, M. Chudy, A. Dybko and Z. Brzozka, *Electrophoresis*, 2016, **37**, 536–544.
- 54 M. W. Kulka, S. Smatty, F. Hehnen, T. Bierewirtz, K. Silberreis, C. Nie, Y. Kerkhoff, C. Grotzinger, S. Friedrich and L. I. Dahms, *Adv. Mater. Interfaces*, 2020, **7**, 2000272.
- 55 Y. Gu, L. Yu, J. Mou, D. Wu, M. Xu, P. Zhou and Y. Ren, *Mar. Drugs*, 2020, **18**, 371.
- 56 Z. Zhao, H. Ni, Z. Han, T. Jiang, Y. Xu, X. Lu and P. Ye, *ACS Appl. Mater. Interfaces*, 2013, **5**, 7808–7818.
- 57 D. Gan, A. Mueller and K. L. Wooley, *J. Polym. Sci., Part A: Polym. Chem.*, 2003, **41**, 3531–3540.